April 29, 2025

Honorable August Pfluger
Chairman
Republican Study Committee
U.S. House of Representatives
Washington, DC 20515

Honorable Bruce Westerman
Chairman
Committee on Natural Resources
U.S. House of Representatives
Washington, DC 20515

*Re: How CBO and Joint Committee Staff Prepare Dynamic Analyses*

Dear Chairman Pfluger and Chairman Westerman:

This letter responds to your questions regarding how the Congressional Budget Office (CBO) and staff of the Joint Committee on Taxation (Joint Committee staff) develop estimates of the budgetary and economic effects of legislation. The agencies have collaborated on the responses to your questions. Joint Committee staff took the lead in responding to the first three questions, and CBO drafted the responses to the last two questions.

**Question 1.** *How do you define a static score, a conventional score, and a dynamic score?*

CBO and Joint Committee staff provide the Members of Congress with estimates of the changes in future outlays and receipts that would result from proposed changes in federal legislation. Neither CBO nor Joint Committee staff produces what are generally understood to be "static estimates." Every estimate of the effects of legislation prepared by CBO and Joint Committee staff includes potential behavioral changes by individuals and business entities in response to the changes in the economic environment that would be created by the proposed legislation.

Our understanding is that when people refer to a "static estimate," they believe that the estimate accounts for absolutely no possible changes in behavior by either individuals or business entities in response to the changes in the economic environment that would be created by the proposed legislation.

Aside from when House Rule XIII requires otherwise, CBO and Joint Committee staff provide the Members of Congress with "conventional estimates" of proposed changes in legislation. A conventional estimate compares projected federal direct spending or receipts under the proposal with projected outlays or receipts under present law—that is, compared with the direct spending and receipts projected under the baseline that CBO prepares in compliance with provisions of the Deficit Control Act. All conventional estimates include potential behavioral changes by individuals and business entities in response to the changes in the economic environment that would be created by the proposed legislation, but they do not assume that such changes would alter the size of the U.S. economy as projected in CBO's macroeconomic baseline. In short, conventional estimates prepared by CBO and Joint Committee staff are not "static"; they reflect myriad microeconomic behavioral responses but do not reflect possible macroeconomic changes.

House Rule XIII requires that in certain circumstances, CBO and Joint Committee staff include macroeconomic behavioral changes in addition to the behavioral responses reflected in conventional estimates. The estimates that include those macroeconomic behavioral changes are commonly referred to as dynamic and are discussed in more detail below.

**Question 1a.** *What factors do you generally consider in a conventional score that are not included in a static score?*

CBO's and Joint Committee staff's models generally assume that individuals and business entities behave rationally and incorporate other behaviors as indicated by data and recent research. Behaviors in conventional estimates include the following:

- Changes in the timing of transactions and income recognition;
- Changes among business sectors or the legal form of doing business;
- Changes in types of portfolio investments;
- Changes in the form in which an employee receives compensation;

- Changes in the amount, types, and timing of consumption; and
- Tax planning and tax avoidance strategies.[1]

**Question 1b.** *Does your conventional analysis always hold GDP fixed?*

Generally, a conventional estimate prepared by CBO or Joint Committee staff does not incorporate a change in the size of the U.S. economy as projected in the CBO baseline of gross national product (GNP). However, if the legislation would significantly change the population of the United States, then the agencies allow for the size of the economy to change by estimating the income of the affected portion of the population and accounting for the budgetary effects of benefits provided to and taxes paid by those people.

**Question 1c.** *What factors in a dynamic score would you generally consider not present in a conventional score?*

Macroeconomic analyses prepared by CBO or Joint Committee staff generally include changes in aggregate labor supply, changes in aggregate domestic investment, total factor productivity, changes in aggregate household saving rates and incomes, national saving, and changes in interest rates, wages, and prices.[2]

**Question 2.** *How do you account for explicit and implicit debt in your dynamic analysis? Specifically, explain how changes in explicit government debt crowd out private investment in your dynamic analysis. Similarly, explain how implicit debt crowds out private investment in your dynamic*

---

[1] For more details and examples of behavior incorporated in Joint Committee staff conventional estimates, see Joint Committee on Taxation, *The Joint Committee on Taxation Revenue Estimating Process* (January 2025), https://tinyurl.com/mr33k3jz. For more information about how CBO prepares cost estimates, see Congressional Budget Office, *CBO Describes Its Cost-Estimating Process* (April 2023), www.cbo.gov/publication/59003, and "Does CBO do 'dynamic analysis'?" (accessed April 25, 2025), www.cbo.gov/faqs#dynamic.

[2] Joint Committee staff recently described its current macroeconomic models and provided an example of the factors that result in differences between an underlying conventional estimate and the estimate that results from a macroeconomic analysis. See Joint Committee on Taxation, *Overview of JCT Methodology for Analyzing the Macroeconomic Effects of Proposed Changes in Tax Law* (December 2024), https://tinyurl.com/5n7h6tvp. Also see Congressional Budget Office, "How the Expiring Individual Income Tax Provisions in the 2017 Tax Act Affect CBO's Economic Forecast" (December 2024), www.cbo.gov/publication/60986.

*analysis. Describe any economic differences across policy changes, including transfer payments, public investment, and tax reductions.*

Increases in explicit government debt due to a policy proposal can direct savings away from private investment toward federal debt service and increase the interest rate that the Federal government has to pay to attract additional funds to finance the budget deficit. The extent to which an increase in government debt can ultimately "crowd out" private investment depends on how people in the United States alter their savings behavior in response to such a change in economic conditions, as well as the response of foreign investors and monetary policy. Each of the three macroeconomic models used by Joint Committee staff accounts for domestic and foreign responses to increases in government debt. The models also incorporate different assumptions about how individuals make their saving decisions. (Those assumptions include expectations about the future and saver versus non-saver households.) For a given policy change, the magnitude of the crowding-out effect can therefore differ across models and ultimately depends on the specific policy under analysis.

In CBO's assessment, increases in the federal deficit reduce private investment. Larger deficits boost aggregate demand and reduce national saving, which in turn raises interest rates and crowds out private investment. The extent to which a policy would affect aggregate demand and national saving depends on its specific details.

Implicit government debt from Social Security and Medicare obligations is relevant for saving behavior in macroeconomic models in which people are assumed to anticipate future changes to federal benefits. In such models, there is a distinction between a payable benefits scenario, in which future benefits are limited to contemporaneous funding after trust fund reserve depletion, and a scheduled benefits scenario, in which future benefits are those specified under current law without regard to the trust fund balances. A payable benefits scenario requires households to forecast an indeterminate reduction in their Social Security benefits at some time in the future. For consistency across all models with varying degrees of individual foresight, Joint Committee staff assumes that scheduled benefits will be paid and that the government's budget is unified. CBO uses a suite of models, including its overlapping generations model, to analyze the effects of changes in debt related to Social Security and other policies.[3]

---

[3] See Congressional Budget Office, *The Long-Term Budget Outlook Under Alternative Scenarios for the Economy and the Budget* (July 2023), pp. 8–10, www.cbo.gov/publication/59233.

**Question 3.** *Provide information about processes for reconciling differences in static and dynamic scoring approaches between the CBO and JCT, especially on legislation with both revenue and expenditure implications.*

When proposed legislation involves policy changes that could alter individuals' or business entities' decisions in ways that would affect federal outlays as well as similar or related policy changes that alter individuals' or business entities' decisions in ways that would affect federal tax receipts, CBO and Joint Committee staff collaborate. Sometimes that collaboration involves specific joint modeling exercises; such is the case when the two agencies provide estimates of changes in health policy. Other times, the collaboration involves using the estimates developed by one agency in the estimates of the other agency. For example, in assessing certain proposals that would change legal immigration and improve tax compliance, Joint Committee staff uses the output of CBO's immigration analysis as the basis for its estimates of the policy's tax provisions. Likewise, CBO uses Joint Committee staff's estimates of the income and payroll tax offset for its analysis of a proposed new fee. In still other situations, the collaboration involves conversations to ensure that CBO and Joint Committee staff incorporate comparable behavioral response parameters when providing estimates to Members who may be weighing options to achieve a policy goal through an outlay program or through a tax expenditure.

CBO's macroeconomic models focus on the details of federal outlays and incorporate tax changes more generally. By contrast, Joint Committee staff's macroeconomic models focus on the details of the Internal Revenue Code and incorporate federal outlays more generally. The two agencies have limited experience conducting collaborative analysis of the effects of macroeconomic changes, as required under House Rule XIII, of legislation that involves significant changes in direct spending programs and significant changes in the Internal Revenue Code.[4] In such situations, the agencies would probably separately estimate macroeconomic changes using existing models and then discuss the results to determine how to report the macroeconomic results to the House of Representatives to fulfill the requirement of House Rule XIII.

**Question 4.** *Describe any differences between how the CBO and JCT would score a bill up for consideration compared to how CBO and JCT*

[4] In 2015, the agencies collaborated on such an analysis. See Congressional Budget Office, cost estimate for H.R. 3762, Restoring Americans' Healthcare Freedom Reconciliation Act of 2015 (October 20, 2015), www.cbo.gov/publication/50918.

*would incorporate the impacts of that bill into the economic baseline if it were signed into law. Please provide details on whether any of those differences are due to restrictions imposed by Congress.*

Analyses from CBO and Joint Committee staff are used by the Congress in several ways. The term "score" typically refers to cost estimates provided to the Budget Committees for budget enforcement purposes.[5] Those advisory estimates provide information that the Congress can use as it implements rules and procedures when considering legislation that would affect the federal budget. Those rules and procedures are determined and enforced by the Budget Committees.

There are three significant ways in which the cost estimate for a bill might differ from the full budgetary consequences reflected in CBO's economic and budgetary baseline after enactment.[6] First, CBO and Joint Committee staff may be directed not to include certain budgetary effects when preparing a cost estimate to be used for budget enforcement. If the bill is enacted, however, the effects that were excluded from the estimate are reflected in the baseline. Second, unless the Congress requires or requests otherwise, cost estimates typically reflect the expectation that the size of the economy remains unchanged. The budgetary effects of the bill incorporated in the baseline reflect the effects of the bill on the size of the economy and on other macroeconomic variables. Finally, cost estimates are made relative to the scoring baseline. If the effects of the bill interact with the economic and legislative changes that have occurred since the scoring baseline was set, the total effect reflected in subsequent baselines will differ from the amount reported in the cost estimate. Two recent reconciliation acts provide concrete examples of those differences.

In the case of the 2022 reconciliation act (Public Law 117-169), all three of the reasons described above contributed to differences between the cost estimate for the act and the effects reflected in subsequent baselines produced by CBO. (Joint Committee staff provided the estimates for changes to tax policy enacted in the 2022 reconciliation act. By statute, Joint Committee staff provides revenue estimates for legislation that would change the Internal Revenue Code.)

---

[5] For more information on CBO's cost estimating process, see Congressional Budget Office, *CBO Explains How It Develops the Budget Baseline* (April 2023), www.cbo.gov/publication/58916.

[6] In addition to those effects, which are specific to the details of the bill, any enacted legislation that changes deficits and debt will alter net interest costs in CBO's subsequent baselines. Those effects on net interest costs are not reflected in the score.

- Guideline 14 directs CBO to exclude any budgetary savings that may arise when mandatory funding is provided for program management or administration. Therefore, the cost estimate for the 2022 act prepared for budget enforcement purposes did not reflect revenue effects from enhanced Internal Revenue Service resources.[7]

- Dynamic estimates were not required or requested for the 2022 reconciliation act. As a result, estimates published by Joint Committee staff and CBO did not reflect any macroeconomic effects of the act.[8]

- Finally, as has been noted by Joint Committee staff, estimates for the 2022 reconciliation act were prepared relative to a 2021 scoring baseline. Economic developments that occurred after that baseline was set would have increased Joint Committee staff's estimated costs of various energy provisions.[9] In addition, the last year of the scoring baseline was 2031, but CBO's subsequent baseline reflected the effects of the act through 2033.

CBO's baseline projections following enactment of the 2022 reconciliation act reflected the estimates prepared by CBO and Joint Committee staff that were used for budget enforcement as well as budgetary effects stemming from the three reasons described above.

Similarly, there were differences between Joint Committee staff's cost estimate for the 2017 tax act (P.L. 115-97) and the effects reflected in CBO's subsequent baseline projections. Those differences were primarily driven by economic and legislative changes that occurred after the scoring baseline was set that affected the costs of the provisions included in the 2017 tax act. For example, CBO's April 2018 baseline projections incorporated revised historical data and changes in the economic outlook

---

[7] See Congressional Budget Office, Estimated Budgetary Effects of H.R. 5376, the Inflation Reduction Act of 2022 (August 2022), www.cbo.gov/publication/58366.

[8] For more information on the economic effects of the 2022 reconciliation act, see Congressional Budget Office, *The Budget and Economic Outlook: 2023 to 2033* (February 2023), Box 2-1, www.cbo.gov/publication/58848.

[9] For example, data that became available after the 2021 baseline was set were consistent with sustained higher inflation than was projected in the 2021 baseline. That increased the estimated cost of certain credits. For further discussion, see Joint Committee on Taxation, *Factors Considered When Estimating the Revenue Effects of the Energy Provisions of Public Law 117-169 and Subsequent Developments* (May 2023), https://tinyurl.com/22wk2mtk.

unrelated to the 2017 tax act that increased output.[10] Such changes in output affect the cost of tax changes. CBO's April 2018 projections reflected the interaction between the 2017 tax act and those economic changes, as well as interactions between the 2017 tax act and legislation enacted in early 2018 that significantly increased spending.

**Question 5.** *Increasing domestic energy production can reduce U.S. energy prices, which would significantly impact every sector of the U.S. economy. Please provide a detailed description of the model and methodology you use to dynamically score the effects of changes in domestic energy production on the U.S. economy and gross domestic product. What assumptions underlie the model, what input variables does your model consider, how are they weighted, and which inputs have the greatest potential to significantly change the model's output?*

To assess the dynamic effects of policies intended to increase energy production—that is, the effects that those policies would be expected to have on the nation's gross domestic product (GDP)—CBO extended a model it previously constructed to evaluate the economic effects of producing oil and natural gas from shale resources.[11] The extended model accounts for additional short-run and long-run effects that stem from proposed policies aimed at increasing energy production. To illustrate the model's framework, CBO assessed the expected economic effects of two alternative scenarios. The first scenario is based on the Energy Information Administration's (EIA's) High Oil and Gas Supply (HOGS) case examined as part of EIA's most recent *Annual Energy Outlook*.[12] Under that scenario, the production of oil and other liquid fuels increases by almost 30 percent in some years over the 2025–2035 period and natural gas production, by 10 percent. The second scenario is based on a proposed policy to expand offshore lease sales of oil and natural gas in which production increases by smaller amounts.[13]

---

[10] For a discussion about how those revisions changed CBO's economic projections, see Congressional Budget Office, "How the 2017 Tax Act Has Affected CBO's GDP and Budget Projections Since January 2017," *CBO Blog* (February 28, 2019), www.cbo.gov/publication/54994.

[11] Congressional Budget Office, *The Economic and Budgetary Effects of Producing Oil and Natural Gas From Shale* (December 2014), www.cbo.gov/publication/49815.

[12] Energy Information Administration, *Annual Energy Outlook 2025* (April 2025), www.eia.gov/outlooks/aeo/.

[13] BRIDGE Production Act of 2023, H.R. 5616, 118th Cong., www.congress.gov/bill/118th-congress/house-bill/5616.

CBO estimates that the first scenario would increase GDP by about $1.3 trillion over 10 years and would reduce the federal deficit by roughly $300 billion relative to CBO's most recent baseline projections. The second scenario would increase GDP by about $10 billion and would reduce the federal deficit by about $2 billion over 10 years. CBO estimates that the second scenario would result in smaller increases in GDP, in part because the changes in energy supplies are much smaller, but also because significant changes in energy production take time. (The EIA scenario assumed that the increases occurred rapidly.) In both scenarios, the increases in GDP stem from an increase in energy production only. CBO's estimates do not incorporate other provisions of any future bills that might interact with those changes in energy production.

**Analytical Method.** CBO's approach to evaluating the effect of changes in domestic oil and natural gas production on GDP is made up of three components. The first reflects the primary investment effects and measures the changes in aggregate demand and changes in the capital stock attributable to increased investment in the oil and natural gas sector. The second component reflects the productivity effects. It measures how the increased supply of oil and natural gas affects the productivity of labor and capital and how it raises GDP over the longer term as labor and capital are allocated to their more efficient uses—in particular, to energy production. The final component reflects the effects of an increased supply of labor and higher savings, which boost output in the longer term. Following the agency's standard method for incorporating longer-term economic effects, CBO phases in part of the primary investment effect (the increase of the capital stock), the productivity effect, and the increase in labor and savings over four years. The short-run change in aggregate demand (part of the primary investment effect) phases out over the same period.

*Primary Investment Effects.* Initial investments in the extraction of new oil and natural gas resources directly stimulate the economy and boost GDP in the short term. In CBO's estimates, the additional investment in oil and natural gas extraction is the amount of investment that would be needed to produce a given increase in oil and natural gas production if the supply of oil and natural gas rose (whether because of a policy change or some other development). Such investment contributes to the total amount of capital in the economy—boosting the ratio of capital to labor each year and increasing GDP in the long term. That increase in investment would crowd out (or reduce) investment in other sectors of the economy, partly offsetting the investment in oil and natural gas. In the first scenario, CBO preserved EIA's assumption that oil and natural gas production would quickly rise

and modeled that the additional investment would take place quickly enough to generate the large increase in production in the first year. When evaluating a proposed policy change, however, CBO would develop its estimates using information about the time between project development and the start of production. For example, in the second scenario, CBO estimated that production would begin about five years after new leases would be put up for auction.

*Productivity Effects.* The main productivity effect stems from the increased productivity of labor and capital resources used to produce the new oil and natural gas resources, measured as the market price of those supplies under the policy minus the average breakeven price of those supplies. Smaller productivity effects stem from gains from reallocating resources from the more costly oil and natural gas supplies produced to the new, less costly supplies and the productivity gains associated with lower prices of oil and natural gas for consumers and businesses.[14]

*Increased Supply of Labor and Greater Savings Stemming From Productivity Effects.* As productivity rises, households have more income to save and invest. That investment boosts the economy's productive capacity and GDP. Higher productivity also boosts wages, encouraging people to work more and also boosting GDP.

Many factors determine the magnitude of the three components, such as the amount of additional energy produced because of the policy, how energy prices are affected, and how sensitive energy use is to changes in those prices. But how quickly investment responds to the new policy determines the sizes of those three effects over the 2025–2035 period. If long lead times were required before additional capital investment could occur, the effects of a policy on GDP would generally happen later, possibly outside the projection period. Conversely, if investment responded more quickly, the short-run increase in demand and the longer-run increases in the capital stock and change in productivity would provide larger effects on GDP during the projection period.

**Model Inputs.** CBO relies on a number of energy market and broader economic inputs to model the component increases in GDP stemming from policies likely to affect energy production. Several components of the effect

[14] For a detailed description of this productivity effect and its components, see Congressional Budget Office, *The Economic and Budgetary Effects of Producing Oil and Natural Gas From Shale* (December 2014), Appendix B, www.cbo.gov/publication/49815.

of increased energy production on GDP depend on how oil and natural gas prices change when new energy supplies enter the market. CBO's estimate of those price changes depends on many factors, such as the expected amount of the new oil and natural gas supplies resulting from the policy; the extent to which those new supplies displace existing, higher-cost energy production; the size of the market to which those new supplies contribute; and the extent to which consumers are willing to purchase oil and natural gas at different prices.

How oil and natural gas production respond to a policy depends on current market conditions, recent trends, and experiences with similar policies, among other factors, and CBO's model accounts for those factors. For instance, in the second scenario, which requires that new lease offerings for offshore oil and natural gas be made available for auction, CBO based its estimate of the expected effect on oil and natural gas production that similar lease offerings have had on the amount and timing of oil and natural gas produced since 2007.

Similarly, how existing oil and natural gas production responds to the entry of lower-cost competing supplies is based on a recent study that examined the factors driving increased production of oil and natural gas on federal lands.[15] The sizes of those markets as well as the responsiveness of consumers in those markets to changes in prices are based on findings of recent scenarios examining the effects of alternative oil and natural gas prices on energy markets and the availability of alternative oil and natural gas resources.[16] Baseline oil and natural gas prices—market prices prior to the proposed policy—are from CBO's most recent economic projections.[17] For the short-run effects, CBO estimated how much investment in oil and natural gas development would be needed to generate the additional oil and natural gas expected to be produced under the new policy. That estimate is consistent with the approach CBO uses to forecast U.S. production of crude oil and petroleum products, which stems from investment in oil and natural gas wells. (If, instead, increased production occurred because existing wells were immediately more productive, as in EIA's HOGS case, less additional

---

[15] Brian C. Prest, *How Much Would Expanding Federal Oil and Gas Leasing Increase Global Carbon Emissions*, Resources for the Future (September 2024), https://tinyurl.com/2fv98ayp.

[16] Energy Information Administration, *International Energy Outlook 2023* (October 2023), www.eia.gov/outlooks/ieo/, and *Annual Energy Outlook 2025* (April 2025) www.eia.gov/outlooks/aeo/.

[17] Congressional Budget Office, *The Budget and Economic Outlook: 2025 to 2035* (January 2025), www.cbo.gov/publication/60870.

investment would be needed, and the productivity effects and increased supply of labor and greater savings from productivity effects would be larger.) Additional investment boosts GDP through increased demand in the short run and in the long run by increasing the stock of capital.

**Dynamic Effects of Scenario 1 (EIA's High Oil and Gas Supply Case).** CBO assessed the effects on GDP of the HOGS case described in the *Annual Energy Outlook 2025*.[18] Under scenario 1, annual oil production would increase by about 4.9 million barrels per day (M bbl/d), on average, between 2025 and 2035, and natural gas production would increase by about 2.7 trillion cubic feet (Tcf), on average.[19] For reference, the United States produced about 20 M bbl/d of oil and other liquid fuels and about 38 Tcf of natural gas in 2024.[20]

Under this scenario, technically recoverable reserves of oil and natural gas—that is, energy resources that can be produced using current technology without regard to cost—are more plentiful and less costly to develop than in EIA's reference case in the following ways:

- Tight oil, tight gas, and shale gas wells are 50 percent more productive relative to EIA's reference case;

- Undiscovered oil and natural gas resources in Alaska and in the offshore continental United States are 50 percent higher relative to EIA's reference case; and

- The rate of technological improvement in drilling and resource development, which lowers the costs of accessing and producing oil and natural gas supplies, is 50 percent faster relative to EIA's reference case.[21]

---

[18] Energy Information Administration, *Annual Energy Outlook 2025* (April 2025) www.eia.gov/outlooks/aeo/.

[19] An increase in crude oil accounts for about two-thirds of the average increase in the amount of oil and other liquid fuels produced; one-third is accounted for by an increase in natural gas liquids, namely butane, ethane, propane, and natural gasoline. Those natural gas liquids are generally included as part of petroleum supplies.

[20] Energy Information Administration, *Monthly Energy Review* (March 2025), www.eia.gov/totalenergy/data/monthly/previous.php.

[21] Energy Information Administration, *Annual Energy Outlook 2023* (April 2023), Appendix: Case Descriptions, www.eia.gov/outlooks/aeo/narrative/#casedescriptons.

In EIA's scenario, the production increase occurs quickly and then stabilizes, requiring only a small change in the number of wells drilled. As a result of the increase in technically recoverable oil and natural gas, in EIA's HOGS scenario production of oil and other liquid fuels would increase by nearly 30 percent in some years and natural gas production by 10 percent over the 2025–2035 period.

In practice, market responses that lead to increased energy production tend to take place more slowly and involve more extensive drilling. Accounting for an increase in drilling, CBO estimates that the resulting increase in oil and natural gas production would raise GDP by about 0.3 percent each year through 2035, on average, or by about $120 billion annually. The combined GDP increase through 2035 would total about $1.3 trillion (see Table 1). In the first year of the projection period, the primary investment effect would account for the entire effect on GDP, and its share of the effect would decline over the next three years before holding roughly steady. In the fifth through tenth years of the scenario, the primary investment effect would account for half of the total increase in GDP. CBO estimates that the

**Table 1.**

## Dynamic Effects of Scenario 1

| | 2025 | 2026 | 2027 | 2028 | 2029 | 2030 | 2031 | 2032 | 2033 | 2034 | 2035 | 2025–2035 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Increase in liquid fuel supply[a] | | | | | | | | | | | | |
| Millions of barrels per day | 3.0 | 3.9 | 4.0 | 4.3 | 4.7 | 5.1 | 5.3 | 5.6 | 5.9 | 6.0 | 6.3 | n.a. |
| Percentage change | 13 | 16 | 17 | 18 | 20 | 21 | 22 | 24 | 25 | 25 | 27 | n.a. |
| Increase in natural gas supply[a] | | | | | | | | | | | | |
| Trillions of cubic feet | 0.8 | 2.5 | 2.9 | 2.9 | 2.5 | 2.6 | 2.6 | 2.3 | 2.9 | 3.6 | 4.2 | n.a. |
| Percentage change | 2 | 7 | 8 | 7 | 6 | 6 | 6 | 5 | 7 | 8 | 10 | n.a. |
| Increase in gross domestic product[b] | | | | | | | | | | | | |
| Billions of dollars | 253 | 15 | 42 | 80 | 99 | 113 | 125 | 134 | 144 | 154 | 164 | 1,324 |
| Percentage change | 0.8 | * | 0.1 | 0.2 | 0.3 | 0.3 | 0.3 | 0.3 | 0.4 | 0.4 | 0.4 | n.a. |

Data source: Congressional Budget Office, using data from the Energy Information Administration.

n.a. = not applicable; * = less than 0.05 percent.

a. Relative to the Energy Information Administration's 2025 reference case.

b. Relative to CBO's January 2025 baseline.

GDP increases would reduce the federal deficit by about $300 billion over the projection period. (The increases in GDP result from changes in energy production only and are evaluated under current law. The change in energy

production does not interact with other effects of any potential policy proposal that could affect energy production in other ways.)

In EIA's analysis of the same case, EIA reports that GDP would increase in inflation-adjusted dollars by about $4.1 trillion through 2035 (equal to about $4.7 trillion in nominal, inflation-unadjusted dollars). That finding is similar to what EIA reported for the same case in its 2022 analysis and somewhat higher than what the agency reported in analyses conducted between 2014 and 2021. By contrast, in 2023, EIA estimated that the same case would increase GDP by $1.1 trillion after 10 years compared with about $360 billion in EIA's 2025 analysis and $560 billion in its 2022 analysis.

**Dynamic Effects of Scenario 2.** CBO also assessed the dynamic effects of H.R. 5616 (a bill introduced by the 118th Congress), which would have required the Bureau of Ocean Energy Management to conduct at least 13 offshore oil and natural gas lease sales on the U.S. Outer Continental Shelf over five years.[22] CBO estimated that the bill would reduce direct spending by $4.2 billion over the 2024–2034 period relative to CBO's February 2024 baseline.[23] That budgetary effect stems primarily from an increase in oil and natural gas production from wells drilled on the newly leased tracts that averaged about 30 M bbl per year (or about 0.1 M bbl/d) for oil and 40 billion cubic feet per year for natural gas over the second half of the 10-year period, reflecting time necessary after leases are sold for energy supplies to be available in the market. Those amounts are each, on average, about 2 percent of the supply increases in the EIA scenario for comparable years. Applying CBO's dynamic modeling—and accounting for the expectation that the increased production on the federal lands would be partially offset by a decrease in domestic production elsewhere—CBO estimates that H.R. 5616 would increase GDP by about $10 billion and would reduce the federal deficit by about $2 billion over the 10-year projection period, with most of those increases taking place in the latter half of the period (see Table 2).

[22] BRIDGE Production Act of 2023, H.R. 5616, 118th Cong., www.congress.gov/bill/118th-congress/house-bill/5616.

[23] Congressional Budget Office, cost estimate for H.R. 5616, BRIDGE Production Act of 2023 (February 28, 2024), www.cbo.gov/publication/60017.

**Table 2.**

## Dynamic Effects of Scenario 2

| | 2024 | 2025 | 2026 | 2027 | 2028 | 2029 | 2030 | 2031 | 2032 | 2033 | 2034 | 2024–2034 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Increase in oil supply[a] | | | | | | | | | | | | |
| Millions of barrels per day | 0 | 0 | 0 | 0 | * | * | 0.02 | 0.05 | 0.08 | 0.11 | 0.12 | n.a. |
| Percentage change | 0 | 0 | 0 | 0 | 0.02 | 0.02 | 0.11 | 0.24 | 0.37 | 0.51 | 0.56 | n.a. |
| Increase in natural gas supply[a] | | | | | | | | | | | | |
| Trillions of cubic feet | 0 | 0 | 0 | 0 | ** | ** | 0.01 | 0.01 | 0.02 | 0.03 | 0.03 | n.a. |
| Percentage change | 0 | 0 | 0 | 0 | † | † | 0.02 | 0.04 | 0.06 | 0.07 | 0.08 | n.a. |
| Increase in gross domestic product[b] | | | | | | | | | | | | |
| Billions of dollars | 0 | 0 | 0 | 0 | ‡ | 0.1 | 0.5 | 1.2 | 2.1 | 2.9 | 3.4 | 10.3 |
| Percentage change | 0 | 0 | 0 | 0 | † | † | † | † | 0.01 | 0.01 | 0.01 | n.a. |

Data source: Congressional Budget Office, using data from the Energy Information Administration.

n.a. = not applicable; * = fewer than 0.01 million barrels per day; ** = less than 0.01 trillion cubic feet; † = less than 0.01 percent; ‡ = less than $0.1 billion.

a. Relative to the Energy Information Administration's 2023 reference case.

b. Relative to CBO's February 2023 baseline.

We hope that this information is helpful to you.

Sincerely,

Phillip L. Swagel
Director
Congressional Budget Office

Thomas Barthold
Chief of Staff
Joint Committee on Taxation

cc: Honorable Jodey C. Arrington
Chairman
House Budget Committee

Honorable Brendan F. Boyle
Ranking Member
House Budget Committee

Honorable Jared Huffman
Ranking Member
Natural Resources Committee